

Elmar Solar Program manual

This manual explains the steps for upgrading the firmware of the Elmar Solar MPPT.

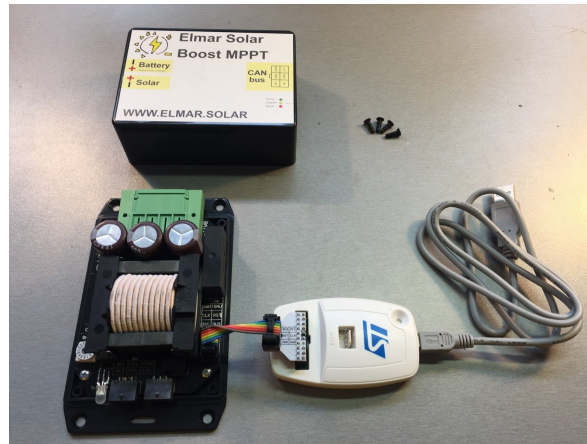
1. Collect all the necessary materials for upgrading the firmware
 - The MPPT of which the firmware will be upgraded
 - A phillips screwdriver
 - A computer running windows
 - ST-LINK/V2 programmer
 - Mini usb cable
 - Elmar Solar 20pin -> 6pin adapter
 - 6 pin ribbon cable



2. Unscrew the 4 screws on the bottom of the MPPT enclosure



3. Connect the ribbon cable, adapter, programmer, and usb cable.



4. Install the USB drivers for the ST programmer available at:

https://www.st.com/content/st_com/en/products/development-tools/software-development-tools/stm32-software-development-tools/stm32-utilities/stsw-link009.html

SLA0047: Image V2 - SOFTWARE LICENSE AGREEMENT 1.18 99.24 KB

TOOLS AND SOFTWARE

Development Tools

HARDWARE DEVELOPMENT TOOLS

Part Number	Manufacturer	Description
STLINKV3SET	ST	STLINK-V3 modular in-circuit debugger and programmer for STM32/STM8

GET SOFTWARE

Part Number	Software Version	Marketing Status	Supplier	Download
STSW-LINK009	2.0.1	Active	ST	Get Software

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5. Install the STM32CubeProg software available at:

https://www.st.com/content/st_com/en/products/development-tools/software-development-tools/stm32-software-development-tools/stm32-programmers/stm32cubeprog.html

Embedded Software

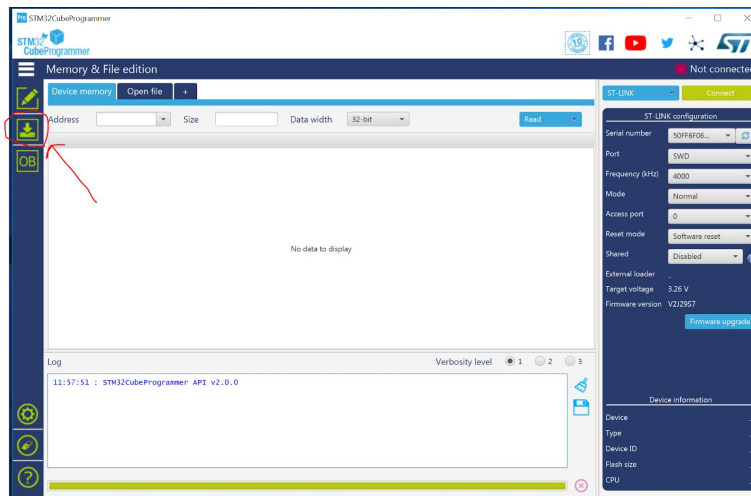
MCU & MPU EMBEDDED SOFTWARE

Part Number	Manufacturer	Description
STM32CubeWB	ST	STM32Cube MCU Package for STM32WB series (HAL and LL low level drivers, USB, File system, RTOS - coming with examples running on ST boards: STM32 Nucleo, Discovery kits and Evaluation boards)
X-CUBE-SBSFU	ST	Secure boot & secure firmware update software expansion for STM32Cube

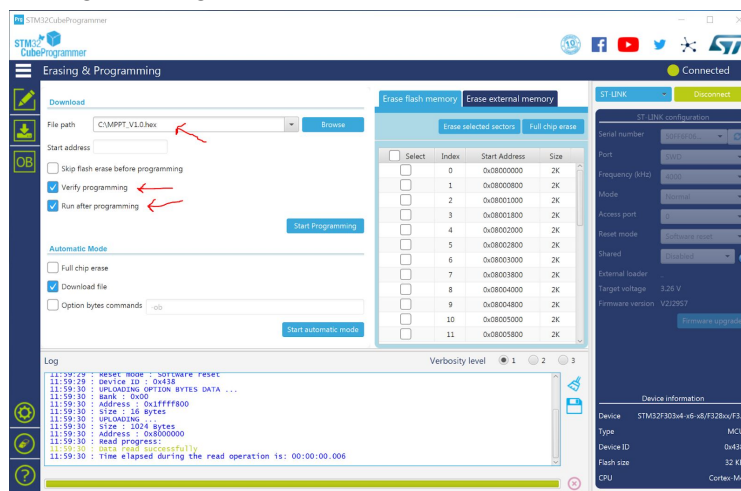
GET SOFTWARE

Part Number	Software Version	Marketing Status	Supplier	Download
STM32CubeProg	2.0.0	Active	ST	Get Software

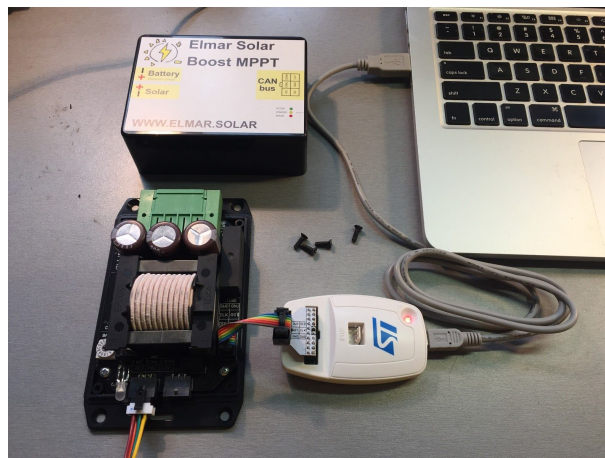
6. Open the STM32CubeProg software, open the “Erasing & Programming” tab



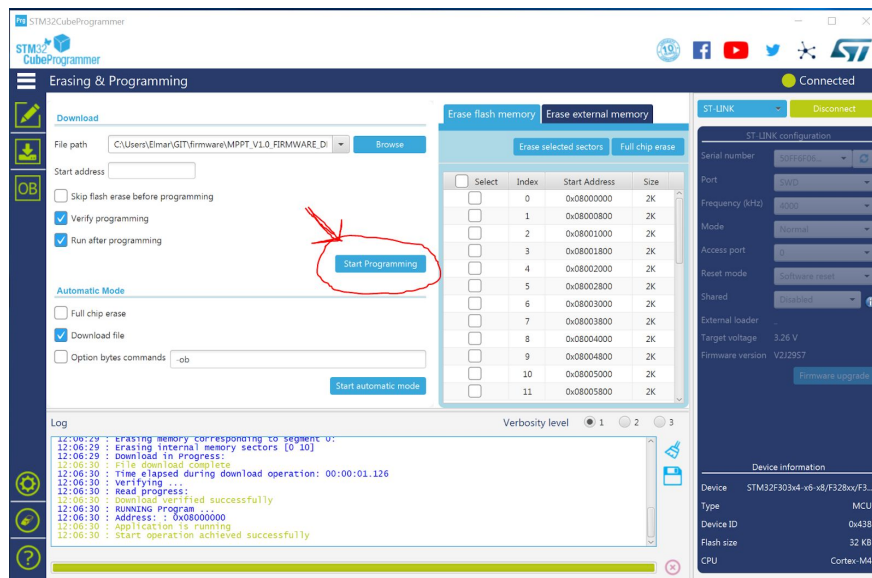
7. Select the location of the firmware file (.hex) at file path, enable “verify programming” and “run after programming”



8. Apply 12V to the CAN connector of the MPPT, plug the USB cable into the computer



9. Hit the “Start Programming” button.



10. Check if the programming was successful

